MUHAMMAD TALHA MASOOD

Full Stack AI Engineer | TypeScript, React & Node.js | PostgreSQL & Python | AI Integration
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PROFILE

Full stack engineer with 4 years building production web applications and integrating AI into real products rather than demos. Works across TypeScript and React on the front, Node.js and Python services on the back, and PostgreSQL underneath, designing schemas, writing the business logic and building the interfaces that sit on top. On the AI side the work is integrating LLM APIs into applications and making the output dependable enough to build on: prompt design, schema constrained structured outputs, validation of what comes back, and handling failures, latency and API cost. Has shipped AI powered classification and analysis features into production, including one now reaching over 10,000 users, and built data pipelines pulling from third party platforms and APIs into structured models that reporting and analysis run on.

TECHNICAL SKILLS

Frontend: TypeScript, React.js, Next.js with App Router, JavaScript (ES6+), responsive and accessible interfaces, state management with Redux and Zustand, data fetching with TanStack Query, form handling and validation with React Hook Form and Zod, Tailwind CSS

Backend: Node.js and Express, server side business logic and REST API design, Python with FastAPI and Django, authentication and authorisation, validation and error handling, background jobs and asynchronous processing with Celery and Redis

Databases: PostgreSQL schema design and relationship modelling, complex queries and aggregation, indexing and query performance optimisation, migrations against live data, Supabase, Drizzle ORM, pgvector, MongoDB, Redis

AI Engineering: Integrating LLMs and AI APIs into production applications (Claude, Gemini, OpenAI, Hugging Face), AI powered classification, analysis and content generation features, structured outputs with prompt design and output validation, embeddings, vector search and RAG pipelines, text and document processing, agents with tool calling and MCP servers, TensorFlow and PyTorch

AI Quality & Cost: Evaluating AI output quality against measured criteria, handling model failures and malformed responses gracefully, managing latency and API cost, iterating on prompts and retrieval against real results rather than assumptions

Integrations & Data: Third party platform and API integration including Meta Ads, authentication flows, error handling and data synchronisation, data pipelines for ingestion and normalisation, structured data modelling for reporting and insight

Practice: Git and collaborative workflows, unit and integration testing, debugging across the stack, code review, technical documentation, Docker, CI/CD pipelines, AWS and Azure, breaking requirements down with product colleagues and estimating work

EXPERIENCE

Full Stack Engineer & Team Lead **Jun 2023 to Present**
Ebryx (Pvt) Ltd. **(Offshore Consultant for Trellix)** *Remote*

- **Full stack delivery:** builds, maintains and improves production web applications in TypeScript, React, Node.js and Python with PostgreSQL behind them, developing backend APIs, database schemas, business logic and the frontend interfaces that consume them.
- **AI integration:** integrates LLM APIs into products to deliver intelligent features, engineering prompts and schema constraints so responses return structured output that downstream code can validate and rely on, alongside RAG pipelines with embeddings and vector search and agents with tool calling.
- **AI output quality:** treats model output as something to be verified rather than trusted, validating responses against expected schemas, handling malformed and failed generations gracefully, and tuning prompts and retrieval against measured results while keeping latency and API cost in view.
- **Data pipelines and integrations:** builds reliable pipelines and integrations with third party platforms and APIs, handling authentication, error cases and data synchronisation, and normalising heterogeneous sources into consistent structured models.
- **Working with structured data:** models and queries structured data so it produces meaningful insight, designing schemas and indexes around the questions being asked and tuning queries to stay responsive as volume grows.
- **Performance and quality:** improves application performance, scalability and reliability by profiling and addressing real bottlenecks, writes tests, reviews code and maintains technical documentation.
- **Collaboration:** works with product and engineering colleagues to break requirements into deliverable work, estimate effort and ship features on time, and leads a small engineering team, reviewing code and mentoring engineers.

Deep Learning Intern **Mar 2022 to Aug 2022**
DataInsight *Islamabad, Pakistan*

- Built and trained models in Python with TensorFlow and PyTorch across text and image processing, handling data preparation and evaluating output quality against measured metrics.

PROJECTS

AI Communication Coach | *TypeScript, React, Node.js, Express, PostgreSQL, Gemini*

- AI product that processes a user's spoken communication and returns a scored, structured analysis with targeted guidance, tracking measured progress across repeated attempts, with separate experiences for learners and their coaches.
- Built end to end: the PostgreSQL schema covering users, sessions and scored transcripts, the Express API, the React and TypeScript interface, and the LLM integration engineered to return schema conformant output that could be validated, persisted and rendered reliably rather than parsed from free form text.

Automated Analysis Product | *Python, ML, NLP, automation*

- AI powered classification feature that analyses submitted content and returns a verdict without manual review, taken from prototype into production and now reaching over 10,000 users as part of their daily workflow, with thresholds tuned so output stayed trustworthy enough to act on.

Skyfeed: Multi Source Data Pipeline | *Python, Django, PostgreSQL*

- Pipeline ingesting from multiple external platforms and APIs on a recurring schedule, normalising heterogeneous formats into one consistent relational model and serving the results through REST APIs, running unattended with retry and failure handling.

Real Time Analytics Dashboard | *React, Node.js, Express, PostgreSQL*

- Full stack dashboard giving users a single interface to monitor, search and filter continuously incoming data in real time, with an indexed schema and tuned queries designed to stay responsive as volume grows.

EDUCATION

B.Sc. in Computer Science, FAST NUCES, Islamabad	2023
FSc (Pre-Engineering), Punjab Group of Colleges, Islamabad	2019

ACHIEVEMENTS

Yearly Star Performer, Ebryx (Pvt) Ltd.	2023
Quarterly Best Performer, Ebryx (Pvt) Ltd.	2024